

Drug tolerance takes shape: persistent homology reveals cyclic cell-state plasticity in EGFR-mutant lung cancer

Hsieh-Ting Lin, M.D.¹ and Yueh-Hua Tu, Ph.D.^{2,*}

¹Department of Medical Oncology, Koo Foundation Sun Yat-Sen Cancer Center, Taipei, Taiwan

²Center for Computational and Systems Biology, National Taiwan University, Taipei, Taiwan

*Correspondence: Yueh-Hua Tu yuehhuatu@ntu.edu.tw

Abstract

Drug-tolerant persister cells drive relapse after EGFR-targeted therapy, yet no scRNA-seq method quantifies the reversibly cyclic state transitions that produce them: standard trajectory tools expose no scalar measure of how robustly cells trace a closed loop in state-space. We present an open-source pipeline that applies persistent homology and Mapper to longitudinal scRNA-seq with permutation-null testing, cell-cycle controls, and gene attribution. Across five EGFR-mutant lung-cancer datasets ($\sim 31,000$ analysed cells), maximum H_1 persistence rose monotonically with osimertinib exposure: PC9 $1.41 \rightarrow 3.78$ ($2.7\times$, $p < 0.01$), PDX YU-006 $2.81 \rightarrow 6.08$ ($2.2\times$), YU-003 $2.79 \rightarrow 3.85$ ($1.4\times$); in a 14-patient EGFR-mutant cohort (Maynard 2020) pooled H_1 grew $5.37 \rightarrow 7.13 \rightarrow 8.05$ across treatment-naïve, persister, and progressive disease, with 3/3 matched pre/post biopsies increasing. The signal survives Tirosh ablation, stringent S/G2M regression, and Harmony batch correction. Loop-associated transcripts include EMT programs in cell line and PDX, with significant system-specific overlap. Max H_1 is a reproducible cross-system phenotype for cell-state plasticity; code is MIT-licensed.

Keywords: topological data analysis; persistent homology; Mapper; single-cell RNA-seq; drug tolerance; EGFR-mutant lung cancer; cell-state plasticity; computational framework.

1 Background

Trajectory methods systematically miss cyclic cell-state structure

Single-cell RNA sequencing has transformed our view of cellular heterogeneity, but the dominant trajectory toolkit — Monocle and Slingshot (tree backbones), PAGA (connectivity graph), and RNA velocity (vector field) — was designed to summarise one-directional progression rather than to quantify reversible inter-state cycling, i.e. cells visiting and revisiting several transcriptional states [1]. PAGA admits non-tree graphs and velocity defines a field, but neither yields a scalar that scores how robustly cells trace a closed loop. It matters in EMT and adaptive drug tolerance [2, 3]. The canonical cell cycle is cyclic in only one direction per cell ($G1 \rightarrow S \rightarrow G2 \rightarrow M$) yet

produces a snapshot H_1 signature indistinguishable from bidirectional inter-state cycling, so we explicitly control for it. Geometrically, inter-state cycling traces a closed loop — a one-dimensional hole that persistent homology detects; directionality is supplied separately by lineage tracing [3].

Topological data analysis as a framework-level remedy

An intuitive picture for biomedical readers. Drug-tolerant cancer cells appear neither to die outright nor to jump into a single resistant state; instead, individual cells visit and revisit *several* transcriptional states over time (fig. 1a). When each cell is plotted as a point in gene-expression space, this inter-state cycling traces a closed loop — the same kind of one-dimensional “hole” that distinguishes a donut from a sphere in topology, and which tree-based trajectory methods discard by force-fitting branching graphs (fig. 1b). Persistent homology quantifies the loop’s robustness across spatial scales: the lifespan of the longest such loop is the *maximum H_1 persistence* statistic used throughout this paper. The headline finding is that this statistic rises monotonically with EGFR-TKI exposure across PC9 cell line, two PDX models, and a 14-patient longitudinal cohort (fig. 1c). Mapper complements this with an interpretable graph in which nodes group transcriptionally similar cells and edges connect related nodes [4].

Formally, persistent homology returns a multiset of birth–death pairs (the persistence diagram) over a filtered sequence of simplicial complexes [5], and Mapper [4] returns a complementary graph summary. Both have been applied individually to bulk cancer transcriptomes [6] and developmental scRNA-seq [7]; prior work has not integrated them into a reproducible, open-source framework for longitudinal scRNA-seq with formal null-model testing and standard controls — the gap our contribution addresses.

Drug-tolerant persisters as a testbed

Drug-tolerant persister cells (DTPs) in EGFR-mutant non-small cell lung cancer (NSCLC) provide an ideal testbed for a topology-based framework. First-line osimertinib gives a median progression-free survival of roughly 18 months, yet essentially every patient relapses from residual disease, and no approved drug specifically targets the persister state [8, 9]. DTPs are a small, slow-growing subpopulation that survives targeted therapy without acquiring mutations [10, 11]. Early work framed DTPs as a single quiescent state [10]; more recent lineage tracing in PC9 osimertinib refines this picture, showing DTPs span multiple transcriptional states and can revert to drug-sensitive phenotypes [3, 12] — the inter-state cycling our framework quantifies, distinct from a single quiescence↔activation switch. Public scRNA-seq datasets cover longitudinal erlotinib and osimertinib treatment at cell-line and PDX scales, but the topological signature has not been formally quantified.

Contribution

We present an open-source pipeline that applies persistent homology (H_0 , H_1) and the Mapper algorithm to longitudinal scRNA-seq, with integrated permutation-null testing, two independent cell-cycle controls, Wasserstein diagram comparison, and Mapper-based gene attribution. We

validate it on five independent EGFR-mutant lung-cancer datasets covering two TKIs (erlotinib, osimertinib), three biological scales (cell line, PDX, patient tumour), and a 14-patient longitudinal cohort. The maximum H_1 persistence rises monotonically with drug exposure across the validation cohorts, survives both cell-cycle controls, and is biologically anchored by EMT-program enrichment in loop-associated transcripts (cell-line and PDX overlap $p = 1.25 \times 10^{-11}$). All analyses are reproducible from raw GEO data via a single Makefile target under MIT license.

2 Results

Erlotinib-treated PC9 cells display robust H_1 persistence (max $H_1 = 3.78$ at day 9)

We first asked whether drug-treated PC9 cells (harboring EGFR exon 19 deletion E746-A750del) exhibit non-trivial H_1 features consistent with cyclic state-space structure. We analyzed GSE134839 [12], a Drop-seq time-series of PC9 cells treated with 2 μ M erlotinib at days 0, 1, 2, 4, 9, and 11 (fig. 2). After standard QC and preprocessing (Methods), 2,942 cells across six timepoints were projected into the top-30 principal components and used to compute the persistence diagrams via the Vietoris–Rips filtration.

All drug-treated timepoints showed non-zero H_1 persistence, with maximum values ranging from 1.08 (D11) to 1.77 (D9) (fig. 3, table 1). Persistence barcodes showed an increased number of long H_1 bars at intermediate and late timepoints. Pairwise Wasserstein distances between persistence diagrams revealed that D0 is separated from all drug-treated timepoints in persistence-diagram space (Wasserstein distance $W = 213$ – 265 between D0 and treated timepoints vs. $W = 26$ – 66 among treated timepoints — roughly an order of magnitude larger), indicating that the distribution of topological features differs most strongly between untreated and treated conditions (fig. 3).

Per-timepoint sample sizes (200–379 cells) left individual permutation tests underpowered (closest D9 $p = 0.098$, D2 $p = 0.126$, near the peak of persister emergence reported by Aissa et al.); the larger validation cohorts below recover $p < 0.01$ (table 2).

Loops persist after Tirosh cell-cycle ablation across five cohorts (ratio 0.71–1.11)

To rule out cell cycle oscillation as a confounder, we removed 39 of 97 Tirosh et al. cell-cycle genes [13] present in our HVG set, re-scaled, and re-computed PCA and persistent homology. The ratio of H_1 after to before cell-cycle ablation ranged from 0.71 to 1.11 across all six timepoints (mean 0.90 ± 0.14), with D11 actually *increasing* after cycle-gene removal (ratio = 1.11) — indicating that the observed cyclic topology is not driven by cell-cycle genes (fig. 3, table 3). This ratio-based test is the standard control in the field and has been interpreted as evidence against cell-cycle artifacts.

(Per-timepoint nulls on the ablated discovery cohort were $p = 0.11$ – 0.58 , expected at the small n further degraded by removing 39 HVGs; larger validation cohorts retain $p < 0.01$; Table S1.)

Stringent S/G2M score regression preserves loops in all five cohorts (ratio 1.07–4.05)

Stringent `sc.pp.regress_out(S_score, G2M_score)` on 1,000-cell subsamples preserves loops across five cohorts (post/pre ratio 1.07–4.05, mean 1.97, every value > 1 ; table 3; full discussion in Supplementary Methods S13). The osimertinib cohort additionally contains zero Tirosh genes in its HVG set, ruling out Tirosh-specific contribution to that cohort by absence.

Osimertinib drives a $2.7\times$ monotonic max H_1 increase over 14 days

On GSE150949 [3], a PC9 lineage-tracing osimertinib time-series (56,419 cells subsampled to 5,000; D0/D3/D7/D14; 4,999 post-QC), max H_1 rose monotonically $1.41 \rightarrow 1.82 \rightarrow 2.67 \rightarrow 3.78$ ($2.7\times$; fig. 4), with permutation tests confirming $p < 0.01$ at D7 and D14 (table 2).

PDX xenografts replicate the osimertinib-driven cycle (YU-006: $2.2\times$, YU-003: $1.4\times$)

In GSE243562 [14] — two EGFR-mutant PDX models (YU-003, YU-006) under matched untreated and osimertinib-residual conditions, 18,874 cells subsampled to 1,500 per group — both models showed an increase in max H_1 under residual disease: YU-003 $2.79 \rightarrow 3.85$ ($1.4\times$, $p < 0.01$); YU-006 $2.81 \rightarrow 6.08$ ($2.2\times$, $p < 0.01$) (fig. 4, table 2). YU-006 matches the cell-line D14 effect; YU-003 is more modest but directionally consistent. This replicates the cell-line finding in an *in vivo* system with stromal, vascular, and host-immune context.

Max H_1 in drug-treated cells approaches patient-tumor levels

To establish a topological reference for untreated human lung adenocarcinoma, we analysed GSE131907 [15] — a 208,506-cell LUAD atlas across 44 patients — subsampling 5,000 primary-tumour epithelial cells (tLung), then 1,500 for VR computation. Treatment-naive patient LUAD shows max $H_1 = 4.77$, substantially higher than treatment-naive PC9 cell line (1.41) or PDX (2.79, 2.81) baselines (fig. 5). The Kim atlas is mixed-driver (EGFR-, KRAS- and other-mutant tumours; not genotype-selected), so this is a generic LUAD reference rather than an EGFR-specific baseline.

Drug treatment moves cell-line and PDX topology from the cell-line-like regime ($H_1 \approx 1\text{--}3$) toward the mixed-driver patient range ($H_1 \approx 4\text{--}5$; fig. 5, closest at osi D14 and YU-006 residual) — a qualitative cross-system comparison confounded by platform and cell count; the load-bearing claim is the within-system monotonic increase (fig. 4).

Timepoint-label permutation null confirms drug-specific signal

The gene-label null tests for non-random topology but not for drug specificity, so we ran a timepoint-label null on osimertinib D14 vs. D0 (drug-exposure annotation shuffled across the pooled 2,499 cells; H_1 recomputed in each permutation). Over 50 permutations, the observed $\Delta H_1 = 2.07$ exceeded the 95th percentile (1.18) in 0/50 trials, giving $p < 0.02$ (Phipson–Smyth +1 correction) — direct evidence that H_1 depends on drug exposure.

EMT genes dominate loop-associated transcripts in both PC9 and PDX systems

What do the H_1 loops represent biologically? To identify the transcriptional programs underlying loop formation, we constructed Mapper graphs on the analyzed cell line data using EMT score as filter function (Methods). The Mapper graph contained 20 nodes and 16 edges; gene-level connectivity scores quantify how strongly each gene varies between Mapper nodes versus globally.

In the cell-line analysis, top-connected genes were canonical EMT markers (*TPM1*, *KRT8*, *KRT19*); MSigDB Hallmark EMT ranked first in top-100 pathway enrichment. The PDX analysis recovered a broader mesenchymal/secretory signature (*S100A4*, *LGALS3*, *COL1A1*) — 3/33 vs. 8/33 Hallmark EMT genes in the cell-line top 100 — consistent with an EMT-like plastic state realised through a partially shared repertoire. Eleven genes overlap between the two top-100 lists (including *KRT19*, *LGALS3*); hypergeometric $p = 1.25 \times 10^{-11}$ against the 13,955-gene background of features expressed in both analyses, $p = 2.1 \times 10^{-4}$ against the more conservative HVG-intersection background (3,000 genes). Differential expression recovered EMT-related signatures in both systems with system-specific second-order pathways (cholesterol homeostasis downregulated in the cell line; estrogen-response upregulated in PDX, a pathway known to interact with EMT via estrogen-receptor-mediated transcription in NSCLC [16]). Full gene-level lists are in Supplementary Tables S2, S2', and S4.

Robustness of gene attribution. Two sensitivity checks support the EMT readout. With PC1 as an unbiased Mapper filter, only 4/33 Hallmark EMT genes appear in the top 100 (vs. 8/33 with the EMT filter) and cell-cycle markers (*MKI67*, *TOP2A*, *HMGB2*) co-rank with EMT markers, so the loops reflect a multi-program state-space rather than a pure EMT signature; PC1↔EMT top-100 overlap is 53/100. Variance-normalised connectivity (dividing each gene's contribution by its global SD) raises Hallmark-EMT hits to 13/33 in the top 100 and drops housekeeping genes out, arguing the enrichment is not an expression-magnitude artefact. Full tables: Supplementary S2 (EMT filter), S2' (PC1 filter), S9 (variance-normalised).

The framework is modular, reproducible, and generalises beyond drug tolerance

The pipeline is fifteen independent modules coordinated by a single `make pipeline` target (Methods; Supplementary Methods S8). The same framework ran without modification on Drop-seq and 10x cell-line data (6,508–56,419 raw cells), 10x PDX data with automated mouse-gene filtering (22,032 cells), and a 208,506-cell patient atlas. Applying it to a new dataset needs only (i) a QC'd and normalised `AnnData` object with a sample/timepoint column, (ii) a Mapper filter (PC1, EMT score, or diffusion component 1 are shipped as defaults), and (iii) a permutation-test subsampling budget (1,000 cells per group is sufficient at $p < 0.05$). We anticipate the framework will generalise to other scRNA-seq perturbation contexts including chemotherapy, immunotherapy escape, and stress-induced dedifferentiation. Resampling stability (CV 8.9–23.5% with ordering preserved), replication under alternative persistence-diagram summaries (total persistence and persistence entropy), and Mapper-parameter robustness (mean Spearman $\rho = 0.994$ across a 4×5 parameter sweep) are reported in Supplementary Methods S10–S12.

The monotonic trend survives Harmony batch correction

To rule out that the observed H_1 growth reflects batch or platform effects rather than intrinsic biology, we applied Harmony batch correction [17] to the top-50 PCA components with dataset-appropriate batch keys (full parameters in Supplementary Methods S7). Max H_1 was then recomputed on the Harmony-corrected embeddings; Figure 6 summarises the comparison.

The osimertinib monotonic trend survives Harmony correction ($1.35 \rightarrow 3.94$; $2.9\times$, vs. $1.71 \rightarrow 3.78$ raw); the YU-006 untreated $\rightarrow$ residual difference is preserved ($2.59 \rightarrow 5.19$ Harmony vs. $2.36 \rightarrow 5.01$ raw); YU-003 becomes more modest but directionally consistent; the Kim LUAD naive baseline is robust (Harmony 4.52 vs. raw 4.77). Full values: Supplementary Table S10. The H_1 growth across drug exposure therefore reflects biology rather than within-dataset batch structure.

In 14 EGFR-mutant NSCLC patients, max H_1 grew monotonically TN $\rightarrow$ PER $\rightarrow$ PD ($5.37 \rightarrow 7.13 \rightarrow 8.05$)

To directly test the framework on clinical tissue, we analysed the full EGFR-mutant subset of Maynard et al. [18]: 13,244 Smart-seq2 cells across 14 patients with biopsies at three treatment stages — treatment-naive (TN, 3,287 cells), persister/residual disease (PER, 4,893 cells), and progressive disease (PD, 5,064 cells). Across 1,500-cell subsamples per stage, pooled max H_1 grew monotonically $5.37 \rightarrow 7.13 \rightarrow 8.05$ (TN-to-PD $1.50\times$; fig. 7a).

Per-patient matched trajectories. Three patients had matched TN + post-treatment biopsies: TH218 ($1.22 \rightarrow 1.65$, $1.35\times$), TH226 ($2.11 \rightarrow 6.86$, $3.25\times$; $\rightarrow 5.01$ PD, $2.37\times$), and AZ003 ($5.32 \rightarrow 8.50$, $1.60\times$; fig. 7a). Post-treatment max H_1 exceeded TN in 3/3 matched pairs (median $1.60\times$, range 1.35 – $3.25\times$; one-sided sign test $p = 0.125$; underpowered at $n = 3$ but directionally consistent; Supplementary Table S11). Raw max H_1 spans ~ 1.2 – 7 across patients at any stage — inter-patient biological and technical heterogeneity — so the within-patient trajectory (always up) is a stronger signal than absolute magnitude, supporting use of max H_1 as a *relative-change* readout.

This patient-cohort result is the most clinically proximal evidence we provide: the same topological signature that rises under drug in PC9 and PDX also rises in real EGFR-mutant NSCLC patient biopsies, in a hypothesis-generating three-of-three matched design.

3 Discussion

A reproducible framework rather than a single dataset finding

The primary contribution of this work is an open-source, modular framework for applying topological data analysis to longitudinal scRNA-seq data, with integrated permutation null testing, cell-cycle ablation, Wasserstein diagram comparison, and Mapper-based gene attribution. The five datasets here span two drugs, three biological scales, and a 14-patient cohort, chosen to stress-test reproducibility rather than to maximise single-cohort significance. This reframes the question from “does drug tolerance produce topological cyclicity?” to “is topological cyclicity

a reproducible cross-system phenotype?” Our answer is yes: $\max H_1$ increases monotonically with osimertinib exposure in three independent systems (PC9 cell line, YU-003 PDX, YU-006 PDX), and all six 1,000-cell permutation tests in table 2 reach $p < 0.05$ (5/6 at $p < 0.01$; the lone exception is the PC9 osimertinib baseline D0 at $p = 0.030$).

Convergent topology, divergent gene programs

A less obvious but biologically informative finding is that the topological signature replicates across cell line and PDX with higher fidelity than the underlying gene programs. Cell-line top-connected genes are dominated by canonical EMT markers (*TPM1*, *KRT8*, *KRT19*); PDX top genes form a broader mesenchymal/secretory signature (*S100A4*, *LGALS3*, *COL1A1*, *AGR2*). The 11-gene overlap between the two top-100 lists is highly significant (hypergeometric $p = 1.25 \times 10^{-11}$) but a minority of either list. We interpret this as evidence that plasticity is realised through partially different gene repertoires depending on microenvironment, consistent with context-dependent EMT programs [19]. The topological statistic abstracts over these specific gene choices, making it a more stable cross-system phenotype than any individual gene signature.

Relationship to existing literature

Our framework complements Oren et al. [3], who used Watermelon lineage tracing to establish PC9 DTP cycling subpopulations, and Aissa et al. [12], who identified discrete DTP states in GSE134839. It detects cyclic behaviour directly from scRNA-seq without lineage-tracing infrastructure, and extends the signal to PDX data. Relative to Rizvi et al. [7] scTDA on mouse ES differentiation, we extend the framework to perturbation-response and add cell-cycle / permutation-null controls absent from the original.

The broader non-genetic drug-tolerance literature — YAP-mediated senescent DTPs [20], melanoma minimal-residual-disease four-state plasticity [21], and the Marine et al. review of non-genetic resistance [22, 23] — provides the biological context. Our contribution against this backdrop is a quantitative, reproducible statistic (topologically invariant in loop existence, metric-dependent in numerical scale; Supplementary Methods S4) for detecting and comparing such structures.

Limitations

Platform and batch confounding. Cross-system comparisons mix Drop-seq (GSE134839) with 10x Chromium (others); the cross-system ordering (cell line < PDX < patient) is reported only qualitatively. Within-system comparisons are platform-matched and survive Harmony batch correction (fig. 6), ruling out within-dataset batch structure as the source. A full scVI cross-platform integration [24] is deferred.

Permutation null specification. The gene-label null tests only for non-random point-cloud structure; the complementary timepoint-label null on osi D14 vs. D0 ($p < 0.02$; Results) directly tests drug dependence.

Patient-cohort validation. The three Maynard matched pre/post pairs (sign-test $p = 0.125$; Results) are hypothesis-generating, not confirmatory; the pooled 14-patient TN→PER→PD monotonic increase is supportive but observational. A prospective paired-biopsy study would be needed to power the within-patient comparison.

Cell-cycle ablation stringency. Tirosh ablation removes 39/97 cell-cycle genes (standard but not exhaustive — quiescence and SASP markers are not in the Tirosh list); the more demanding S/G2M-score regression (ratio 1.07–4.05 across five cohorts) and the PC1-filter Mapper (cell-cycle genes co-rank with EMT markers) together indicate the topology is multi-program rather than purely EMT, and is not a cell-cycle artefact.

Implications and outlook

If drug tolerance operates through topologically cyclic plasticity, state-trapping combinations that collapse the cycle may outperform single-state-targeting strategies. The bidirectional nature of inter-state cycling further provides a conceptual basis for the partial clinical efficacy of TKI rechallenge after a drug holiday in EGFR-mutant NSCLC: cells that survived prior treatment can revert to drug-sensitive states. The cross-scale reproducibility of the signature argues that $\max H_1$ is a quantitative phenotype for cell-state plasticity, suitable for benchmarking drug responses across labs and likely to generalise to adjacent perturbation-response settings.

Why might cyclic plasticity be adaptive? We speculate that a robust 1-cycle corresponds to a Waddington-style limit cycle around a shallow basin [25, 26], making cyclic plasticity adaptive under fluctuating selection as bet-hedging [27, 28]; in this view osimertinib both selects pre-existing cycles and induces new ones, although distinguishing the two would require lineage tracing or a controlled-clonal-bottleneck design. Adaptive therapy [29, 30] and state-trapping combinations become testable hypotheses against this readout.

4 Conclusions

Drug treatment does not simply push lung-cancer cells along a one-way road to resistance: it reshapes the cell-state landscape into a form that admits inter-state cycling across EMT-like programs, consistent with the cycling subpopulations established by lineage tracing in PC9 [3]. The maximum H_1 persistence introduced here quantifies how robust that cycling becomes, rises monotonically with osimertinib across cell-line and PDX validation cohorts, survives two independent cell-cycle controls and Harmony batch correction. Loop-associated genes are enriched for EMT programs in both cell line and PDX when an EMT-aware Mapper filter is used; an unbiased PC1 filter additionally surfaces cell-cycle and proliferation genes, indicating the topological loops are multi-program rather than purely EMT. We expect the readout to generalise to other reversible-plasticity settings where standard trajectory methods fall short.

5 Methods

Datasets

GSE134839 [12]: PC9 cells treated with 2 μ M erlotinib; six Drop-seq DGE matrices (D0, D1, D2, D4, D9, D11); 6,508 total cells. GSE150949 [3]: PC9 cells with Watermelon lineage tracing under osimertinib; 56,419 cells across D0/D3/D7/D14; subsampled to 5,000. GSE243562 [14]: two EGFR-mutant PDX models (YU-003, YU-006) under osimertinib; 22,032 cells across four samples (untreated and residual disease for each model). GSE131907 [15]: treatment-naive patient LUAD atlas; 208,506 cells across 44 patients; subsampled to 5,000 tLung epithelial cells, then a further 1,500 cells for tractable Vietoris–Rips computation (filter and seed details in Supplementary Methods S1).

Preprocessing

We used scanpy 1.11 [31] with standard QC (`min_genes = 200`, `max_genes = 5000`, `max_pct_mito = 20%`, `min_cells = 3`), total-count normalisation to 10,000 counts per cell, log1p transformation, 3,000 highly-variable genes, regression of total counts and mitochondrial percentage, scaling to max 10, and PCA with 50 components (`random_state = 42`). HVG batch keys per cohort: `timepoint` for GSE134839 and GSE150949, `pdx` for GSE243562, and `Sample` for GSE131907 (rationale in Supplementary Methods S2). For PDX data, mouse genes were removed by a symbol-pattern heuristic (Supplementary Methods S3; 29% of features filtered; limitations in Discussion). Cell-cycle scores were computed from Tirosh et al. [13] S- and G2/M-phase gene lists via `sc.tl.score_genes_cell_cycle`.

Persistent homology

We used `ripser` 0.6 [32, 33] to compute the Vietoris–Rips filtration up to H_1 with the Euclidean metric on the top-30 PCA coordinates; homology is computed over $\mathbb{F}_2$ coefficients (ripser default). Maximum persistence is defined as $\max(\text{death} - \text{birth})$ over finite H_1 bars; this is a hybrid statistic whose existence is a topological invariant but whose numerical scale is metric-dependent (Supplementary Methods S4). Wasserstein distances between diagrams were computed with `persim` 0.3 using the 2-Wasserstein metric. The p -Wasserstein distance between persistence diagrams is stable under bounded perturbations of the input point cloud given a finite-degree- p total-persistence hypothesis [34, 35].

Principal component count sensitivity

Max H_1 depends on the number of PCs retained: across $n_{\text{PCs}} \in \{10, 20, 30, 50\}$ absolute values vary within a factor of two, but the drug-treated $>$ untreated ordering is preserved at every PC count. The $n_{\text{PCs}} = 30$ default follows prior scTDA work [7]. Full sweep numbers and the rationale for the hybrid-statistic framing are in Supplementary Methods S4–S5 and Supplementary Table S5.

Permutation tests

For each group, we shuffled expression values within each cell (permuting gene labels independently for every cell), re-fit PCA using `sklearn`, and recomputed H_1 max persistence. The p -value is the fraction of null permutations with $\max H_1 \geq$ the observed value. Discovery cohort used 500 permutations per timepoint; validation cohorts used 100 permutations on 1,000-cell subsamples for tractability.

Mapper

We used `kmapper` 2.0 [36] with an auto-tuned DBSCAN clusterer (`eps` set to the 50th percentile of 10th-nearest-neighbor distances in each cover element). Default parameters were $n_{\text{cubes}} = 15$ and `perc_overlap` = 0.3. Parameter sensitivity was assessed across $n_{\text{cubes}} \in \{10, 15, 20, 25, 30\}$ and `overlap` $\in \{0.2, 0.3, 0.4, 0.5\}$.

EMT score (used as Mapper filter for the cell-line analysis) was computed via `scanpy.tl.score_genes` on a 33-gene MSigDB Hallmark EMT cassette (28 of 33 present in GSE134839; full gene list in Supplementary Methods S6).

Gene connectivity

For each gene g and Mapper node N with cells C_N , we computed $|\text{mean}(g \mid C_N) - \text{mean}(g \mid \text{all cells})| \times |C_N|$, summed over all nodes, and divided by total cells. Higher scores indicate genes that vary more between Mapper nodes than globally.

Pathway enrichment

Gene set enrichment was performed with `gseapy` 1.1 against MSigDB_Hallmark_2020 and GO_Biological_Process_2021.

Computational environment

All analyses ran on Python 3.11 in a conda environment specified in `envs/environment.yml`; random seed 42 is used for every stochastic operation. Full package versions and platform notes (macOS arm64 and Linux x86_64): Supplementary Methods S14.

Pipeline architecture and unit tests

The framework is organised as fifteen numbered modules (`01_preprocess.py` through `15_maynard_full_cohort`) that each accept a standard `AnnData` object and produce typed intermediate outputs. Users can substitute their own datasets by providing an `AnnData` file with a `timepoint` observation column. Full module list and the `pytest` unit-test suite (8 tests, all passing) are described in Supplementary Methods S8–S9.

Code availability

Code is available at <https://github.com/htlin222/sctda-cancer-plasticity> (MIT license). The version-tagged software release accompanying this submission is `v2.7.1-submission`,

archived as a public GitHub release at <https://github.com/htlin222/sctda-cancer-plasticity/releases/tag/v2.7.1-submission>. The complete pipeline is reproducible via `make pipeline` from raw GEO data to all figures and tables. A unified `sctda` CLI is planned for v2.0; the current pipeline is invoked via the Makefile and individual `scripts/OX_*.py` files. A `CITATION.cff` file at the repository root provides machine-readable citation metadata for software-citation tooling.

Data availability

All raw scRNA-seq data used in this study are publicly available from the Gene Expression Omnibus under accessions GSE134839, GSE150949, GSE243562, and GSE131907; the Maynard et al. 2020 EGFR-mutant cohort was obtained from the authors' public Google Drive archive referenced in Maynard et al. [18]. Processed AnnData objects, persistence diagrams, and Mapper graphs are regenerable end-to-end from the raw GEO data by running `make pipeline`; seeds and parameters are pinned in the repository so derived outputs are bit-stable across runs.

Declarations

Ethics approval and consent to participate

Not applicable — all analyses use previously published, de-identified public datasets.

Consent for publication

Not applicable.

Availability of data and materials

All raw data and processed intermediate files are available as described in the Methods section (Data availability). Supplementary tables accompanying this manuscript: S1 (cell-cycle ablation permutation p -values), S2 (EMT-filter Mapper gene connectivity), S2' (PC1-filter Mapper gene connectivity), S3 (Mapper parameter sweep), S4 (PDX gene attribution), S5 (Max H_1 vs principal component count), S6 (functional persistence summaries: max H_1 , total persistence, persistence entropy across ten cohorts), S7 (subsample variance: mean $\pm$ SD of max H_1 across five random seeds for four cohorts), S8 (Mapper parameter-sweep Spearman rank correlations), S9 (variance-normalized Mapper gene connectivity), S10 (Harmony batch-correction raw vs. corrected max H_1 comparison), S11 (Maynard 2020 full EGFR-mutant cohort per-patient max H_1 and within-patient trajectory statistics (14 patients, 3 matched pre/post pairs)), S-stringent-CC (stringent cell-cycle regression ratios; reproduced in the lower block of Table 3).

Competing interests

The authors declare that they have no competing interests.

Funding

This work received no specific external grant funding and was carried out using the authors' institutional computing resources at Koo Foundation Sun Yat-Sen Cancer Center (H.-T.L.) and the Center for Computational and Systems Biology, National Taiwan University (Y.-H.T.).

Authors' contributions

H.-T.L. (first author) and Y.-H.T. (corresponding author) jointly conceived the study and designed the analytical strategy. H.-T.L. implemented the pipeline, performed all analyses, produced all figures, and wrote the original draft (CRediT: Conceptualisation, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Visualisation). Y.-H.T. supervised the project, contributed methodological and validation-strategy review, and revised the manuscript (CRediT: Conceptualisation, Methodology, Supervision, Writing – review & editing, Project administration). Both authors read and approved the final manuscript.

Acknowledgements

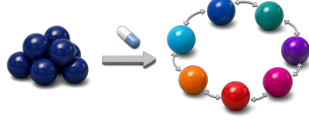
The authors thank the contributors of GSE134839 [12], GSE150949 [3], GSE243562 [14], GSE131907 [15], and the Maynard et al. 2020 cohort [18] for making their data publicly available, and the open-source developers of scanpy, ripser, persim, kmapper, harmonypy, and giotto-tda.

Figures

Drug-induced cell-state cycling, detected as a topological loop and measured by persistent homology

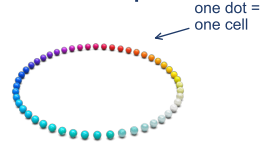
(a) drug pressure induces cycling → (b) scRNA-seq sees it as a loop → (c) persistent homology measures the loop's lifespan

a · biology



Drug pressure makes cancer cells cycle between states

b · scRNA-seq



scRNA-seq sees the cycle as a closed loop



quantify with persistent homology

c · how we measure the loop

How long the loop survives
($\max H_1$ persistence)

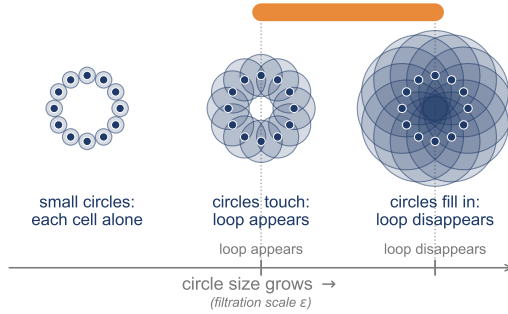


Figure 1: Drug-induced cell-state cycling, detected as a topological loop and measured by persistent homology. (a) Untreated cancer cells occupy a uniform transcriptional state; under EGFR-TKI pressure they diversify into multiple states with reversible transitions between them (cells coloured by state). (b) In scRNA-seq, this reversible cycling appears as a closed loop in cell-state space (one dot = one cell). (c) Persistent homology measures the loop. Around each cell we grow a ball; as the ball radius (the filtration scale ε) increases, the loop is born when balls touch their neighbours and dies when balls fill the loop's interior. The lifespan of this loop — the lifetime of the longest H_1 bar — is the *maximum H_1 persistence* statistic used throughout this work. The mug-and-donut topological-equivalence analogy that motivates the use of homology is described in the Background. Cross-cohort fold-changes are in fig. 5.

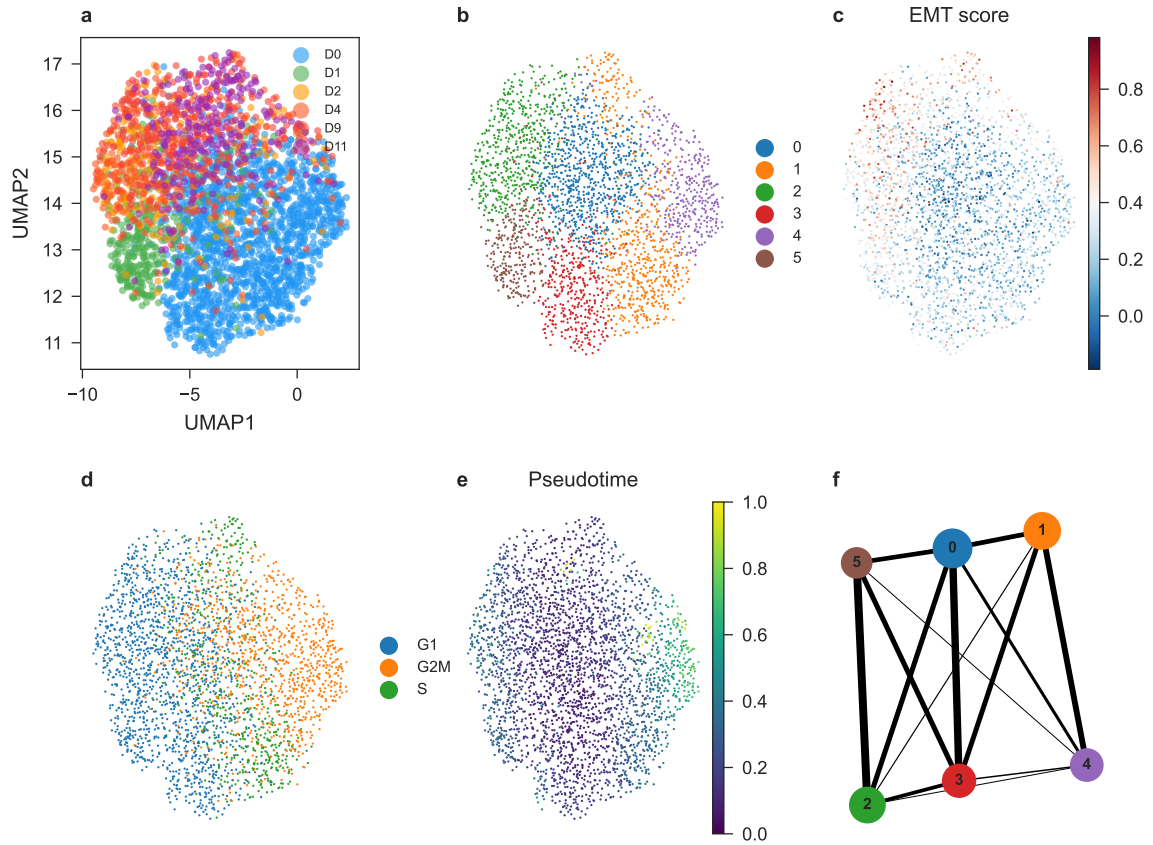
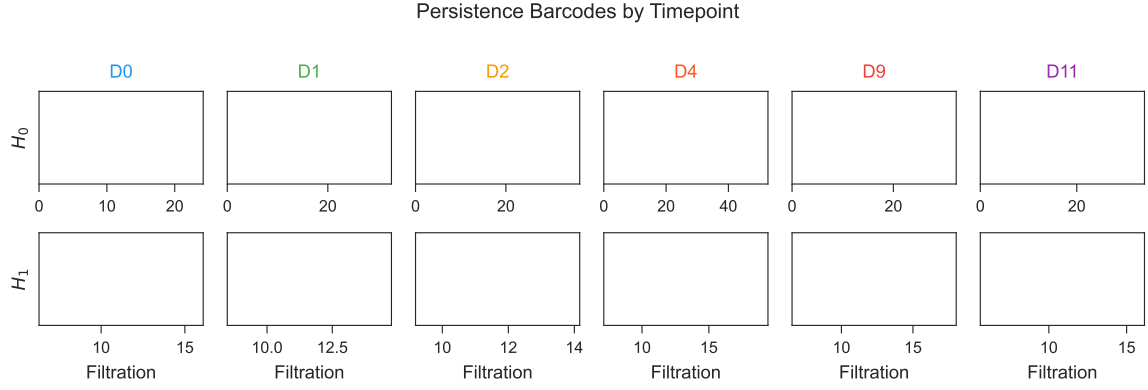
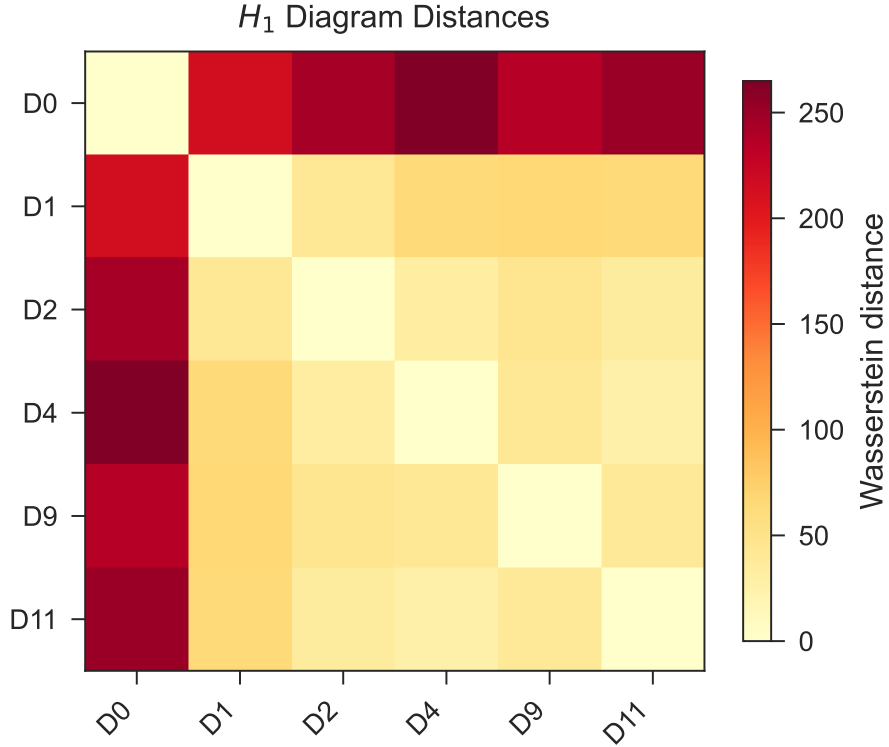


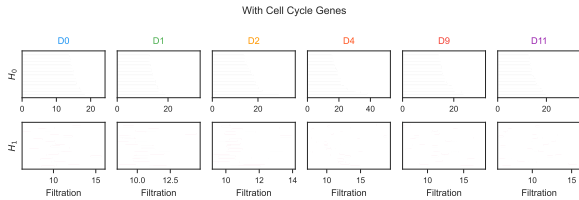
Figure 2: **Study design and cell-line baseline analysis.** (a) UMAP of PC9 erlotinib time course colored by timepoint. (b) Leiden clusters. (c) EMT score. (d) Cell cycle phase. (e) Diffusion pseudotime. (f) PAGA graph. $n = 2,942$ cells across six timepoints.



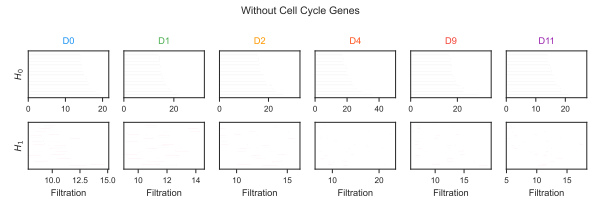
(a) Persistence barcodes by timepoint (top: H_0 ; bottom: H_1).



(b) Pairwise Wasserstein distances between per-timepoint H_1 persistence diagrams.

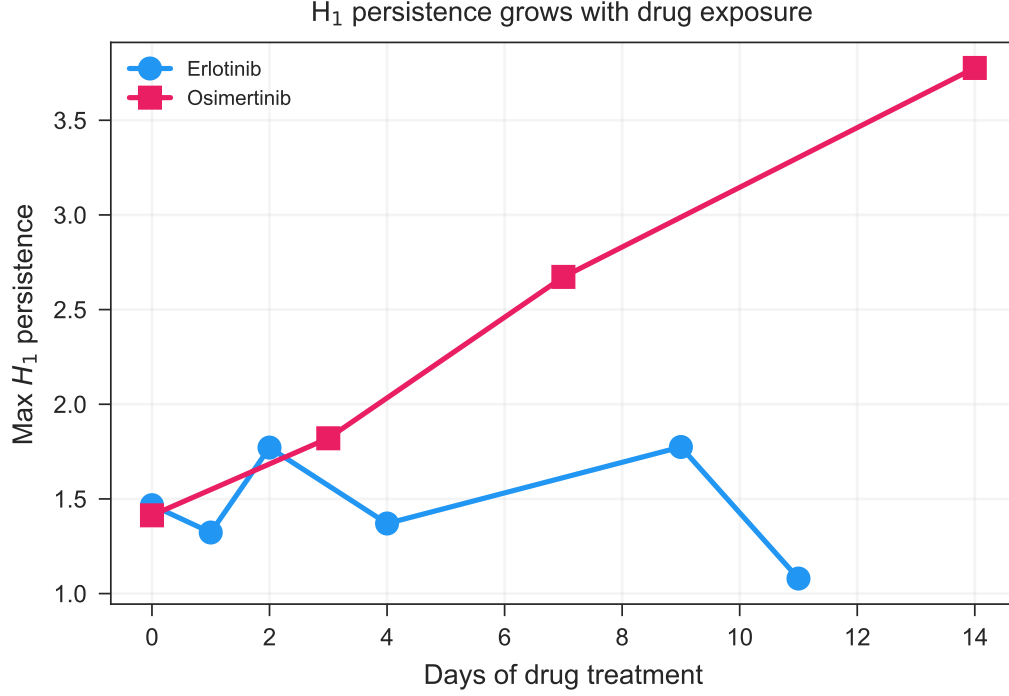


(c) Barcodes with cell-cycle genes.

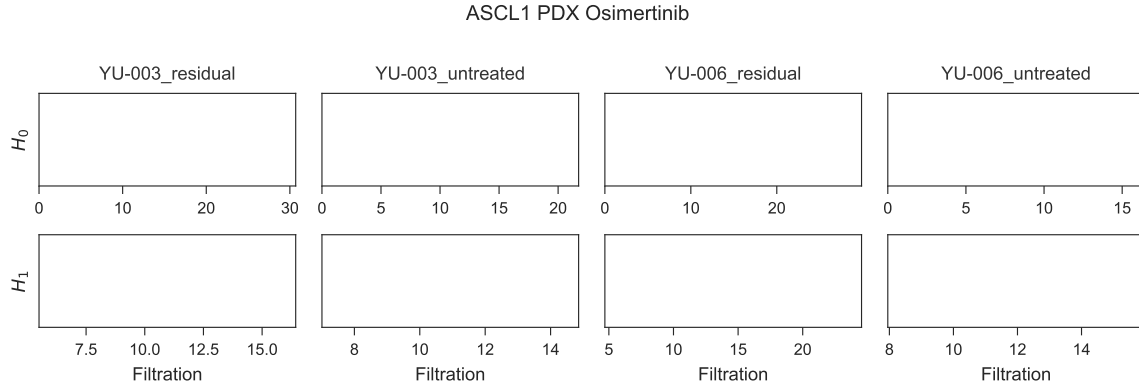


(d) Barcodes without cell-cycle genes.

Figure 3: Persistent homology of the PC9 erlotinib discovery cohort and Tirosh cell-cycle control. (a) Per-timepoint H_0 (top) and H_1 (bottom) persistence barcodes for the GSE134839 discovery cohort; permutation tests' closest approaches to significance are D9 ($p = 0.098$) and D2 ($p = 0.126$). (b) Pairwise Wasserstein distance heatmap between per-timepoint H_1 persistence diagrams: D0 is separated from all drug-treated timepoints in persistence-diagram space. (c, d) H_1 barcodes recomputed with and without 39 Tirosh cell-cycle genes ablated from the HVG set; the persistence-ratio ≥ 0.71 at every timepoint indicates loops are not cell-cycle artifacts.



(a) Max H_1 vs. days for the PC9 erlotinib and osimertinib time courses; osimertinib shows a $2.7\times$ monotonic increase from D0 to D14.



(b) Persistence barcodes for the YU-003 and YU-006 PDX models under untreated and osimertinib-residual conditions; both PDX models increase H_1 persistence under residual disease.

Figure 4: Drug-validation cohorts: osimertinib in PC9 and in EGFR-mutant patient-derived xenografts. (a) PC9 max- H_1 trajectory under erlotinib vs. osimertinib (GSE150949). (b) PDX YU-003 and YU-006 untreated vs. residual barcodes (GSE243562). Together these are the within-system monotonic-increase replications behind the cross-cohort summary in fig. 5.

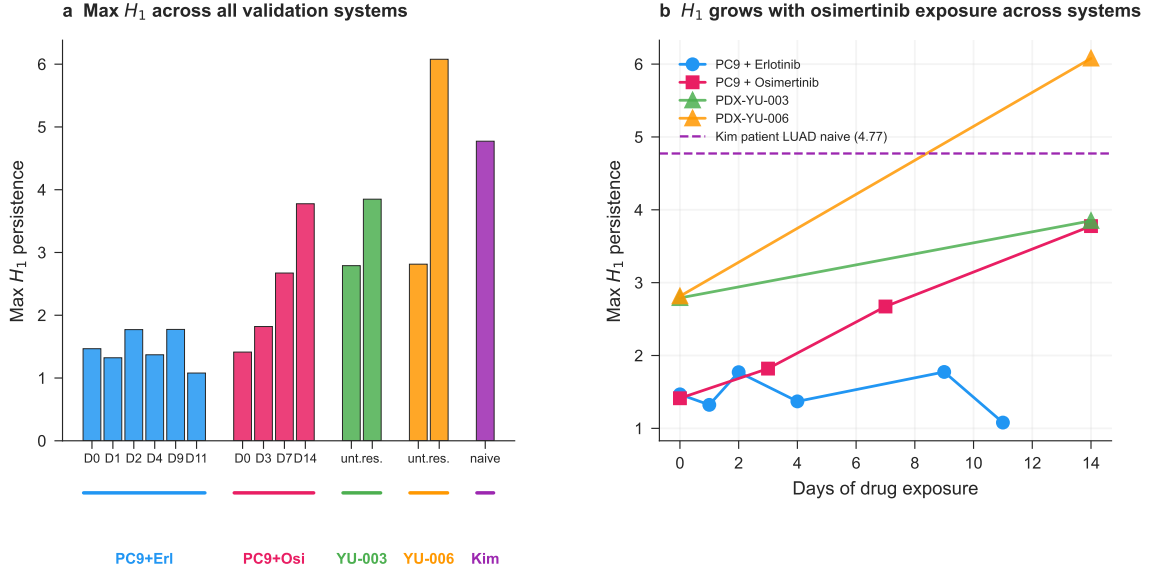


Figure 5: **Master comparison across all biological systems.** (a) Max H_1 bar chart across all conditions. (b) Max H_1 vs. days for all drug-treated systems, with patient-naive baseline as a reference line. Drug treatment pushes cell line and PDX topology toward patient-tumor-like complexity.

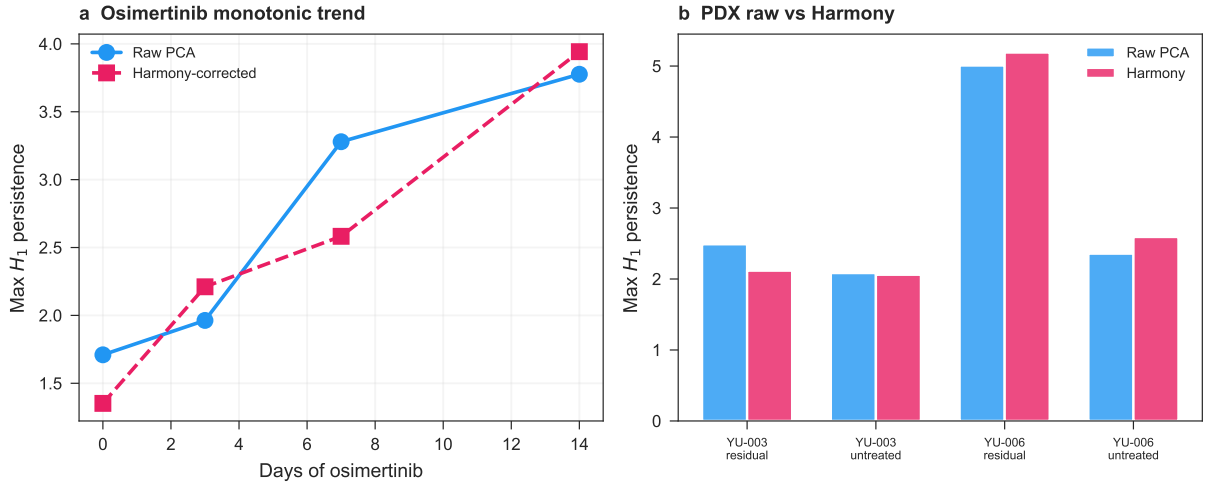
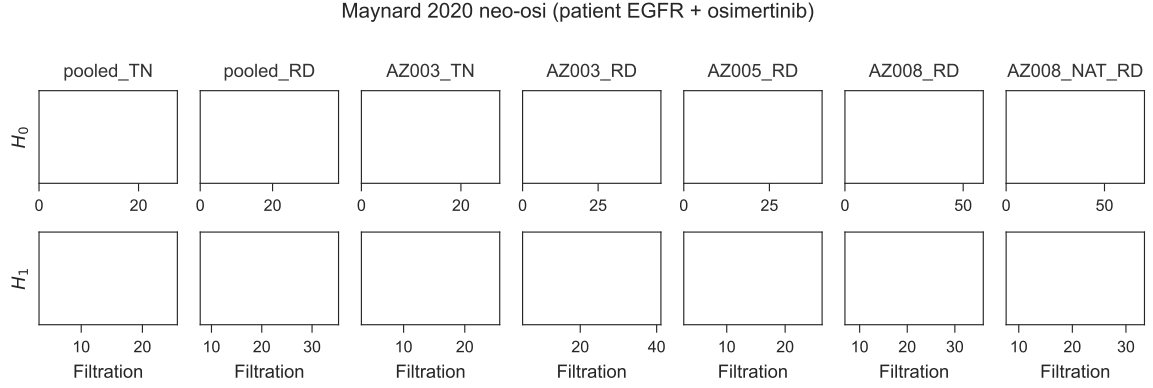
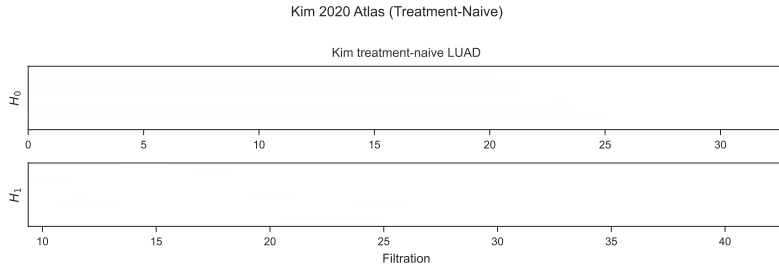


Figure 6: **Harmony batch correction sensitivity.** (a) Max H_1 vs. days of osimertinib exposure in the PC9 cell line cohort (GSE150949), computed on raw PCA (blue) and on Harmony-corrected embeddings (pink, dashed). Both show monotonic growth from D0 to D14. (b) Max H_1 per PDX group (untreated vs. residual disease) in GSE243562, raw (blue) vs. Harmony-corrected (pink). The YU-006 pre/post effect is preserved after batch correction.



(a) Maynard 2020 EGFR-mutant longitudinal cohort (14 patients, 13,244 cells): persistence barcodes (H_0 top, H_1 bottom) pooled across treatment-naïve (TN), persister (PER), and progressive disease (PD) stages. Pooled max H_1 : $5.37 \rightarrow 7.13 \rightarrow 8.05$; three of three matched pre/post biopsies increase post-treatment (median $1.60\times$).



(b) Treatment-naïve lung adenocarcinoma reference from the Kim et al. 2020 atlas (GSE131907, $n = 1,500$ epithelial cells subsampled from 5,000 tLung cells): persistence barcodes establishing the patient-tumor max $H_1 = 4.77$ reference used in fig. 5.

Figure 7: Patient-cohort validation across two independent scRNA-seq studies. (a) Maynard 2020 14-patient EGFR-mutant longitudinal cohort: max H_1 rises monotonically TN $\rightarrow$ PER $\rightarrow$ PD with 3/3 matched pre/post pairs increasing. (b) Kim 2020 LUAD atlas: persistence barcodes for the treatment-naïve primary-tumor epithelium reference. See Supplementary Table S11 for full per-patient Maynard trajectories.

Tables

Table 1: **Per-timepoint H_1 summary (GSE134839, erlotinib)**. Permutation p -values from 500 within-cell gene-label shuffles.

Timepoint	n cells	Max H_1	# H_1 features	Perm. p (500)
D0	1560	1.47	1393	0.394
D1	324	1.32	255	0.552
D2	230	1.77	124	0.126
D4	200	1.37	54	0.144
D9	379	1.77	204	0.098
D11	249	1.08	124	0.266

Table 2: **Validation cohort H_1 permutation tests (100 permutations, 1,000-cell subsamples)**. Values from 1,500-cell subsample used in main analysis; permutation tests used 1,000-cell subsamples (see Methods). Per-cohort max H_1 values vary across subsample sizes (1,000 vs. 1,500 cells) due to the strong subsample dependence of persistent homology on cell count (Supplementary Table S7, CV 8.9–23.5%); values reported in Table 2 use the 1,000-cell permutation-test subsample for internal consistency with the p -value column. Note that PC9 osimertinib D0 max H_1 is reported as 1.41 in the main analysis (5,000-cell subsample) and 1.74 here (1,000-cell permutation subsample); these are different subsamples of the same underlying data and are not inconsistent.

Dataset	Group	Observed H_1	p -value
PC9 + Osimertinib	D0	1.74	0.030
PC9 + Osimertinib	D7	3.08	< 0.01
PC9 + Osimertinib	D14	3.78	< 0.01
PDX YU-006	Untreated	3.62	< 0.01
PDX YU-006	Residual	6.08	< 0.01
PDX YU-003	Residual	3.85	< 0.01

Table 3: **Cell-cycle ablation controls.** Top: Tirosh cell-cycle gene removal from HVG set (39 of 97 Tirosh genes present; full discovery-cohort timepoints from GSE134839). Ratio is max H_1 after / before ablation; ratios ≥ 0.71 indicate loops are not cell-cycle artifacts. Bottom: stringent `sc.pp.regress_out` of S-score and G2M-score on 1,000-cell subsamples across five cohorts. Ratio is max H_1 after regression / original; every ratio > 1 indicates the H_1 signal is preserved after removing the dominant cell-cycle variance axis.

Group	n cells	H_1 before	H_1 after	Ratio
<i>Tirosh cell-cycle gene ablation (GSE134839 erlotinib)</i>				
D0	1560	1.47	1.35	0.92
D1	324	1.32	1.28	0.97
D2	230	1.77	1.57	0.89
D4	200	1.37	0.97	0.71
D9	379	1.77	1.39	0.78
D11	249	1.08	1.20	1.11
<i>Stringent S/G2M regression (1,000-cell subsamples)</i>				
Erl D0	1000	1.27	2.85	2.25
Erl D9	379	1.77	1.89	1.07
Erl D11	249	1.08	4.37	4.05
PDX YU-006 untreated	1000	2.89	3.67	1.27
PDX YU-006 residual	1000	4.65	5.96	1.28

References

- [1] Wouter Saelens, Robrecht Cannoodt, Helena Todorov, and Yvan Saeys. A comparison of single-cell trajectory inference methods. *Nature Biotechnology*, 37:547–554, 2019. doi: 10.1038/s41587-019-0071-9.
- [2] Sydney M. Shaffer, Margaret C. Dunagin, Stefan R. Torborg, Eduardo A. Torre, Benjamin Emert, Clemens Krepler, Marilda Beqiri, Katrin Sproesser, Patricia A. Brafford, Min Xiao, Elisabeth Eggan, Ioannis N. Anastopoulos, Cesar A. Vargas-Garcia, Abhyudai Singh, Katherine L. Nathanson, Meenhard Herlyn, and Arjun Raj. Rare cell variability and drug-induced reprogramming as a mode of cancer drug resistance. *Nature*, 546:431–435, 2017. doi: 10.1038/nature22794.
- [3] Yaara Oren, Michael Tsabar, Michael S. Cuoco, Liat Amir-Zilberstein, Heidie F. Cabanos, Jan-Christian Hutter, Bomiao Hu, Pratiksha I. Thakore, Marcin Tabaka, Charles P. Fulco, William Colgan, Brandon M. Cuevas, Sara A. Hurvitz, Dennis J. Slamon, Amy Deik, Kerry A. Pierce, Clary Clish, Aaron N. Hata, Elma Zaganjor, Galit Lahav, Katerina Politi, Joan S. Brugge, and Aviv Regev. Cycling cancer persister cells arise from lineages with distinct programs. *Nature*, 596:576–582, 2021. doi: 10.1038/s41586-021-03796-6.
- [4] Gurjeet Singh, Facundo Mémoli, and Gunnar Carlsson. Topological methods for the analysis of high dimensional data sets and 3D object recognition. In *Eurographics Symposium on Point-Based Graphics*, pages 91–100, 2007.
- [5] Herbert Edelsbrunner and John Harer. *Computational Topology: An Introduction*. American Mathematical Society, 2010. ISBN 978-0-8218-4925-5.
- [6] Monica Nicolau, Arnold J. Levine, and Gunnar Carlsson. Topology based data analysis identifies a subgroup of breast cancers with a unique mutational profile and excellent survival. *Proceedings of the National Academy of Sciences USA*, 108:7265–7270, 2011. doi: 10.1073/pnas.1102826108.
- [7] Abbas H. Rizvi, Pablo G. Camara, Elena K. Kandror, Thomas J. Roberts, Ira Schieren, Tom Maniatis, and Raul Rabadan. Single-cell topological RNA-seq analysis reveals insights into cellular differentiation and development. *Nature Biotechnology*, 35:551–560, 2017. doi: 10.1038/nbt.3854.
- [8] Suresh S. Ramalingam, Johan Vansteenkiste, David Planchard, Byoung Chul Cho, Jhanelle E. Gray, Yuichiro Ohe, Caicun Zhou, Thanyanan Reungwetwattana, Ying Cheng, Busyamas Chewaskulyong, Riyaz Shah, Manuel Cobo, Ki Hyeong Lee, Parneet Cheema, Marcello Tiseo, Thomas John, Meng-Chih Lin, Fumio Imamura, Takayasu Kurata, Alexander Todd, Rachel Hodge, Matilde Saggese, Yuri Rukazenzov, and Jean-Charles Soria. Overall survival with osimertinib in untreated, egfr-mutated advanced nscl. *New England Journal of Medicine*, 382(1):41–50, 2020. doi: 10.1056/NEJMoa1913662.
- [9] Zofia Piotrowska, Matthew J. Niederst, Chris A. Karlovich, Heather A. Wakelee, Joel W. Neal, Mari Mino-Kenudson, Linnea Fulton, Aaron N. Hata, Elizabeth L. Lockerman, Anuj

- Kalsy, Subba Digumarthy, Alona Muzikansky, Mitch Raponi, Aurora R. Garcia, Hillary E. Mulvey, Madeleine K. Parks, Richard H. DiCecca, Dora Dias-Santagata, A. John Iafrate, Alice T. Shaw, Andrew R. Allen, Jeffrey A. Engelman, and Lecia V. Sequist. Heterogeneity underlies the emergence of egfr^{T790M} -mediated resistance to osimertinib. *Cancer Discovery*, 8(12):1529–1539, 2018. doi: 10.1158/2159-8290.CD-18-1022.
- [10] Sreenath V. Sharma, Diana Y. Lee, Bihua Li, Margaret P. Quinlan, Fumiyuki Takahashi, Shyamala Maheswaran, Ultan McDermott, Nancy Azizian, Lee Zou, Michael A. Fischbach, Kwok-Kin Wong, Kathleyn Brandstetter, Ben Wittner, Sridhar Ramaswamy, Marie Classon, and Jeff Settleman. A chromatin-mediated reversible drug-tolerant state in cancer cell subpopulations. *Cell*, 141:69–80, 2010. doi: 10.1016/j.cell.2010.02.027.
- [11] Aaron N. Hata, Matthew J. Niederst, Hannah L. Archibald, Maria Gomez-Caraballo, Faria M. Siddiqui, Hillary E. Mulvey, Yosef E. Maruvka, Fei Ji, Hyo-eun C. Bhang, Viveksagar Krishnamurthy Radhakrishna, Giulia Siravegna, Haichuan Hu, Sana Raoof, Elizabeth Lockerman, Anuj Kalsy, Dana Lee, Celina L. Keating, David A. Ruddy, Leah J. Damon, Adam S. Crystal, Carlotta Costa, Zofia Piotrowska, Alberto Bardelli, Anthony J. Iafrate, Ruslan I. Sadreyev, Frank Stegmeier, Gad Getz, Lecia V. Sequist, Anthony C. Faber, and Jeffrey A. Engelman. Tumor cells can follow distinct evolutionary paths to become resistant to epidermal growth factor receptor inhibition. *Nature Medicine*, 22:262–269, 2016. doi: 10.1038/nm.4040.
- [12] Alexandre F. Aissa, A. B. M. M. K. Islam, Majd M. Ariss, Camille C. Go, Alexandra E. Rader, Ryan D. Conrardy, Alexa M. Gajda, Carlota Rubio-Perez, Klara Valyi-Nagy, Mary Pasquinelli, Lawrence E. Feldman, Stefan J. Green, Nuria Lopez-Bigas, Maxim V. Frolov, and Elizaveta V. Benevolenskaya. Single-cell transcriptional changes associated with drug tolerance and response to combination therapies in cancer. *Nature Communications*, 12: 1628, 2021. doi: 10.1038/s41467-021-21884-z.
- [13] Itay Tirosh, Benjamin Izar, Sanjay M. Prakadan, Marc H. Wadsworth, Daniel Treacy, John J. Trombetta, Asaf Rotem, Christopher Rodman, Christine Lian, George Murphy, Mohammad Fallahi-Sichani, Ken Dutton-Regester, Jia-Ren Lin, Ofir Cohen, Parin Shah, Diana Lu, Alex S. Genshaft, Travis K. Hughes, Carly G. K. Ziegler, Samuel W. Kazer, Aleth Gaillard, Kellie E. Kolb, Alexandra-Chloé Villani, Cory M. Johannessen, Aleksandr Y. Andreev, Eliezer M. Van Allen, Monica Bertagnolli, Peter K. Sorger, Ryan J. Sullivan, Keith T. Flaherty, Dennie T. Frederick, Judit Jané-Valbuena, Charles H. Yoon, Orit Rozenblatt-Rosen, Alex K. Shalek, Aviv Regev, and Levi A. Garraway. Dissecting the multicellular ecosystem of metastatic melanoma by single-cell RNA-seq. *Science*, 352:189–196, 2016. doi: 10.1126/science.aad0501.
- [14] Bomiao Hu, Marc Wiesehöfer, Fernando J. de Miguel, Zongzhi Liu, Lok-Hei Chan, Zongyu Zhou, Jessie Sotelo, Habeeb Alsudani, Alexandra Nottingham, Zenta Walther, Robert Homer, Daniel J. Boffa, William W. Lockwood, Dana S. Neel, Anna Wurtz, Christine M. Lovly, Nicholas Giustini, Can Cai, Anne Chiang, Yuanquan Huang, Elisa de Stanchina, Catherine B. Meador, Jessica A. Hellyer, Aaron N. Hata, Daniel A. Haber, and Katerina Politi. ASCL1

- drives tolerance to osimertinib in EGFR mutant lung cancer in permissive cellular contexts. *Cancer Research*, 84(8):1303–1319, 2024. doi: 10.1158/0008-5472.CAN-23-0438.
- [15] Nayoung Kim, Hong Kwan Kim, Kyungjong Lee, Yourae Hong, Jae Ho Cho, Jung Won Choi, Jung-Il Lee, Yeon-Lim Suh, Bo Mi Ku, Hye Hyeon Eum, Soyeon Choi, Yoon-La Choi, Je-Gun Joung, Woong-Yang Park, Hyun Ae Jung, Jong-Mu Sun, Se-Hoon Lee, Jin Seok Ahn, Keunchil Park, Myung-Ju Ahn, and Hae-Ock Lee. Single-cell RNA sequencing demonstrates the molecular and cellular reprogramming of metastatic lung adenocarcinoma. *Nature Communications*, 11:2285, 2020. doi: 10.1038/s41467-020-16164-1.
- [16] Jill M. Siegfried and Laura P. Stabile. Estrogenic steroid hormones in lung cancer. *Seminars in Oncology*, 41(1):5–16, 2014. doi: 10.1053/j.seminoncol.2013.12.009.
- [17] Ilya Korsunsky, Nghia Millard, Jean Fan, Kamil Slowikowski, Fan Zhang, Kevin Wei, Yuriy Baglaenko, Michael Brenner, Po-Ru Loh, and Soumya Raychaudhuri. Fast, sensitive and accurate integration of single-cell data with Harmony. *Nature Methods*, 16(12):1289–1296, 2019. doi: 10.1038/s41592-019-0619-0.
- [18] Ashley Maynard, Caroline E. McCoach, Julia K. Rotow, Lincoln Harris, Franziska Haderk, D. Lucas Kerr, Elizabeth A. Yu, Erin L. Schenk, Weilun Tan, Alexander Zee, Michelle Tan, Philippe Gui, Tasha Lea, Wei Wu, Anatoly Urisman, Kirk Jones, Rene Sit, Pallav Kumar Kolli, Erin Seeley, Yaron Gesthalter, Daniel D. Le, Kevin A. Yamauchi, David M. Naeger, Sourav Bandyopadhyay, Khyati Shah, Lauren Cech, Nicholas J. Thomas, Anshal Gupta, Mayra Gonzalez, Hien Do, Lisa Tan, Bianca Bacaltos, Rafael Gomez-Sjoberg, Matthew Gubens, Collin M. Blakely, Julia K. Rotow, Norma F. Neff, Robert C. Doebele, Jonathan Weissman, Jonathan W. Riess, Quan-Thang Le, Spyros Darmanis, and Trever G. Bivona. Therapy-induced evolution of human lung cancer revealed by single-cell RNA sequencing. *Cell*, 182:1232–1251.e22, 2020. doi: 10.1016/j.cell.2020.07.017.
- [19] Anushka Dongre and Robert A. Weinberg. New insights into the mechanisms of epithelial-mesenchymal transition and implications for cancer. *Nature Reviews Molecular Cell Biology*, 20:69–84, 2019. doi: 10.1038/s41580-018-0080-4.
- [20] Kari J. Kurppa, Yao Liu, Ciric To, Tinghu Zhang, Mengyang Fan, Amir Vajdi, Erik H. Knelson, Yangsheng Xie, Klothilde Lim, Paloma Cejas, et al. Treatment-induced tumor dormancy through YAP-mediated transcriptional reprogramming of the apoptotic pathway. *Cancer Cell*, 37(1):104–122.e12, 2020. doi: 10.1016/j.ccell.2019.12.006.
- [21] Florian Rambow, Aljosja Rogiers, Oskar Marin-Bejar, Sara Aibar, Julia Femel, Michael Dewaele, Panagiotis Karras, Daniel Brown, Young Hwan Chang, Maria Debiec-Rychter, et al. Toward minimal residual disease-directed therapy in melanoma. *Cell*, 174(4):843–855.e19, 2018. doi: 10.1016/j.cell.2018.06.025.
- [22] Jean-Christophe Marine, Sarah-Jane Dawson, and Mark A. Dawson. Non-genetic mechanisms of therapeutic resistance in cancer. *Nature Reviews Cancer*, 20(12):743–756, 2020. doi: 10.1038/s41568-020-00302-4.

- [23] Shuai Shen, Sebastien Vagner, and Caroline Robert. Persistent cancer cells: the deadly survivors. *Cell*, 183(4):860–874, 2020. doi: 10.1016/j.cell.2020.10.027.
- [24] Romain Lopez, Jeffrey Regier, Michael B. Cole, Michael I. Jordan, and Nir Yosef. Deep generative modeling for single-cell transcriptomics. *Nature Methods*, 15(12):1053–1058, 2018. doi: 10.1038/s41592-018-0229-2.
- [25] Sui Huang. The molecular and mathematical basis of Waddington’s epigenetic landscape: a framework for post-Darwinian biology? *BioEssays*, 34(2):149–157, 2012. doi: 10.1002/bies.201100031.
- [26] Angela Oliveira Pisco and Sui Huang. Non-genetic cancer cell plasticity and therapy-induced stemness in tumour relapse: ‘What does not kill me strengthens me’. *British Journal of Cancer*, 112(11):1725–1732, 2015. doi: 10.1038/bjc.2015.146.
- [27] Amy Brock, Hannah Chang, and Sui Huang. Non-genetic heterogeneity – a mutation-independent driving force for the somatic evolution of tumours. *Nature Reviews Genetics*, 10(5):336–342, 2009. doi: 10.1038/nrg2556.
- [28] Piyush B. Gupta, Christine M. Fillmore, Guozhi Jiang, Sagi D. Shapira, Kai Tao, Charlotte Kuperwasser, and Eric S. Lander. Stochastic state transitions give rise to phenotypic equilibrium in populations of cancer cells. *Cell*, 146(4):633–644, 2011. doi: 10.1016/j.cell.2011.07.026.
- [29] Robert A. Gatenby. A change of strategy in the war on cancer. *Nature*, 459(7246):508–509, 2009. doi: 10.1038/459508a.
- [30] Pedro M. Enriquez-Navas, Yoonseok Kam, Tuhin Das, Sameera Hassan, Ariosto Silva, Parastou Foroutan, Epifanio Ruiz, Gary Martinez, Susan Minton, Robert J. Gillies, and Robert A. Gatenby. Exploiting evolutionary principles to prolong tumor control in preclinical models of breast cancer. *Science Translational Medicine*, 8(327):327ra24, 2016. doi: 10.1126/scitranslmed.aad7842.
- [31] F. Alexander Wolf, Philipp Angerer, and Fabian J. Theis. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biology*, 19:15, 2018. doi: 10.1186/s13059-017-1382-0.
- [32] Ulrich Bauer. Ripser: efficient computation of Vietoris–Rips persistence barcodes. *Journal of Applied and Computational Topology*, 5(3):391–423, 2021. doi: 10.1007/s41468-021-00071-5.
- [33] Christopher Tralie, Nathaniel Saul, and Rann Bar-On. Ripser.py: a lean persistent homology library for Python. *Journal of Open Source Software*, 3(29):925, 2018. doi: 10.21105/joss.00925.
- [34] David Cohen-Steiner, Herbert Edelsbrunner, and John Harer. Stability of persistence diagrams. *Discrete & Computational Geometry*, 37(1):103–120, 2007. doi: 10.1007/s00454-006-1276-5.
- [35] David Cohen-Steiner, Herbert Edelsbrunner, John Harer, and Yuriy Mileyko. Lipschitz

- functions have l_p -stable persistence. *Foundations of Computational Mathematics*, 10(2): 127–139, 2010. doi: 10.1007/s10208-010-9060-6.
- [36] Hendrik Jacob van Veen, Nathaniel Saul, David Eargle, and Sam W. Mangham. Kepler Mapper: a flexible Python implementation of the Mapper algorithm. *Journal of Open Source Software*, 4(42):1315, 2019. doi: 10.21105/joss.01315.